

Closed-Form Coaxial Non-Premixed Burner Design Tool

Abstract

The objective of our work is to develop a Python-based design tool for a coaxial non-premixed burner capable of performing analytical burner sizing and automatically generating the corresponding nozzle geometry. The main motivation behind this work is to replace manual design calculations with a computational framework that rapidly determines burner dimensions while satisfying the required combustion, flow, and flame stability constraints.

The first part of the work focuses on burner sizing using closed-form analytical equations. Based on the specified thermal power, fuel type, and equivalence ratio, the program calculates the fuel and air flow rates, determines the required fuel tube dimensions from continuity, selects the nearest commercially available tube size, and identifies a common casing diameter satisfying the prescribed mass-flux ratio. The design is subsequently verified by evaluating Reynolds number, Froude number, pressure drop, lift-off margin, Mach number, and operating conditions for both methane and propane. Since methane exhibits the lower lift-off velocity, it governs the final burner sizing, and the resulting burner dimensions are based on this limiting operating condition.

The second part of the work deals with automatic nozzle profile generation. A MATLAB reference implementation based on fifth-order polynomial geometry was first studied to understand the mathematical formulation used for smooth nozzle contour generation. The same methodology was subsequently implemented in Python while using the burner dimensions obtained from the analytical design calculations. Independent polynomial profiles were generated for the fuel inner wall, fuel tube outer wall, casing inner wall, and outer casing wall, ensuring continuous slope, smooth curvature, and positive wall thickness throughout the contraction.

To facilitate computer-aided design, the generated nozzle profiles are automatically exported as separate coordinate files corresponding to each wall of the burner geometry. These coordinate datasets provide the geometric definition required for constructing the burner components in SolidWorks using spline-based curves and revolve features, thereby eliminating manual profile creation and improving dimensional consistency between analytical design and CAD implementation. By which, we updated the current full jet geometry in the SolidWorks.

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1. Introduction

Coaxial non-premixed burners are widely used in combustion research due to their simple configuration and their ability to provide controlled fuel-air mixing. The burner geometry, including the fuel tube, annular air passage, and nozzle profile, significantly influences the flow characteristics, flame stability, and overall burner performance. Therefore, accurate burner sizing and nozzle design are important steps in the development of a reliable experimental burner.

In this work, a Python-based computational tool has been developed to automate the burner sizing process and generate the corresponding nozzle profile. The burner dimensions are obtained using analytical design equations, while the nozzle profile is generated using a quintic polynomial formulation. The mathematical methodology used for the nozzle profile was understood by studying a reference MATLAB implementation. After the nozzle profile is generated, the program automatically exports the geometric coordinates of each burner wall as separate CSV files. These coordinate files can be directly used for constructing the burner geometry in CAD software such as SolidWorks, thereby providing a seamless transition from analytical design to three-dimensional modelling without manual recreation of the nozzle profile.

2. Objectives

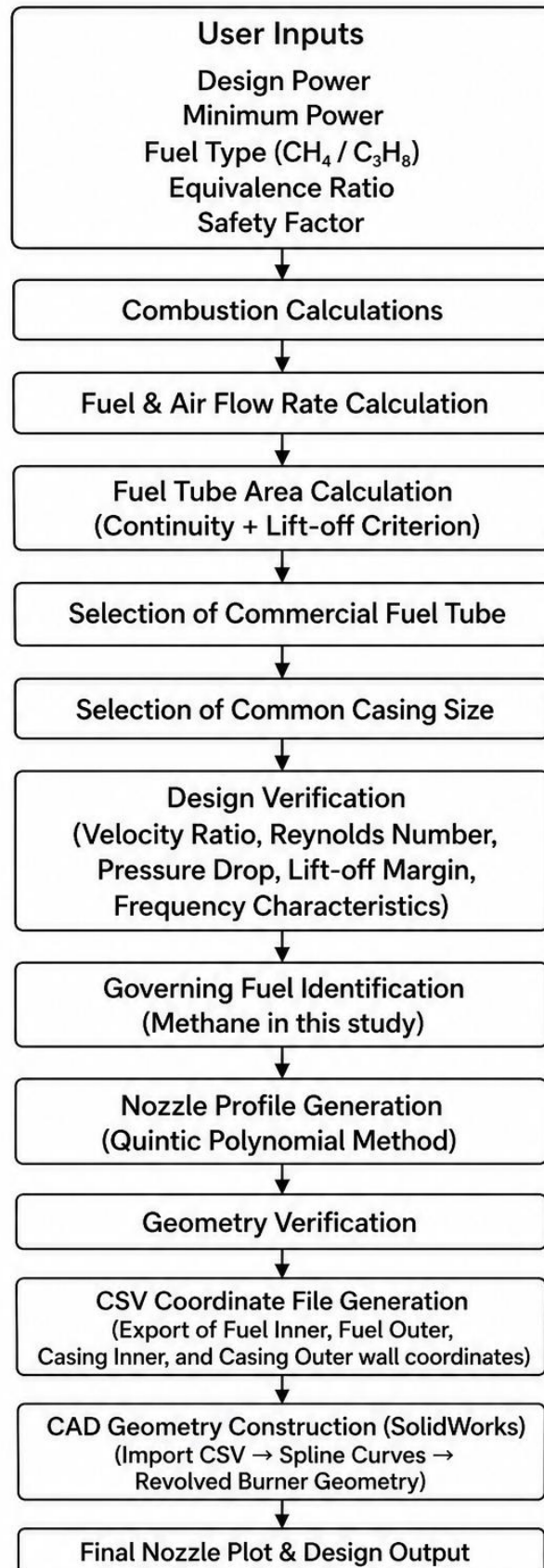
The objective of the present work is to develop a Python-based computational tool for the analytical design of a coaxial non-premixed burner. The program aims to automate the burner sizing process by calculating the required fuel and air flow rates, selecting suitable commercial fuel tubes and casing dimensions, and verifying the final design using relevant engineering parameters. Another important objective is to generate a smooth converging nozzle profile based on the calculated burner dimensions using a fifth-order polynomial. To achieve this, the nozzle profile generation methodology adopted in the reference MATLAB implementation was carefully studied and independently implemented in Python so that it could be integrated with the burner sizing calculations. The developed tool is intended to provide a systematic and flexible design framework that can be further refined and extended for future numerical and experimental investigations.

In addition, the developed tool automatically exports the generated nozzle profiles as separate coordinate files corresponding to the fuel inner wall, fuel tube outer wall, air annulus inner wall, and outer casing wall. These coordinate files preserve the analytically generated geometry and enable direct reconstruction of the burner profile in CAD software such as SolidWorks. This feature eliminates the need for manual geometry creation and provides a consistent link between analytical burner design and subsequent three-dimensional modelling.

3. Overall Workflow

The developed Python program automates the design of a coaxial non-premixed burner from the specified operating conditions. The required fuel and air flow rates are first calculated using combustion relations, after which the fuel tube and common casing dimensions are selected based on continuity, lift-off stability, and commercially available sizes. The obtained burner geometry is then verified using engineering parameters such as velocity ratio, Reynolds number, pressure drop, and frequency characteristics. Finally, methane, being the governing fuel with the lowest lift-off velocity, is used to generate a single converging nozzle profile using a quintic polynomial formulation, and the resulting geometry is plotted for further analysis.

After obtaining the final burner dimensions, the program generates the nozzle profile using a fifth-order polynomial formulation. The generated geometry is then exported as separate coordinate files for each burner wall, enabling direct construction of the burner geometry in CAD software such as SolidWorks for further design and visualization.



4. Burner Design Methodology

The burner dimensions were obtained using a closed-form design procedure. Starting from the required thermal power and fuel properties, the fuel and air flow rates were first calculated using combustion stoichiometry. The governing fuel was then selected based on the lift-off criterion, which determined the allowable design fuel velocity. Using the continuity equation, the required fuel exit area was calculated and matched to the nearest commercially available tube size. A common casing diameter was then selected by satisfying the required mass-flux-ratio limits for both methane and propane. Finally, the obtained design was verified over the complete operating range from 2 kW to 5 kW.

4.1 Design Inputs

The design calculations were performed using the input parameters listed in Table 4.1.

Table 4.1:

Parameter	Value
Design thermal power	5 kW
Minimum operating power	2 kW
Fuel	Methane and Propane
Equivalence ratio, ϕ	1.0
Target velocity ratio, V_f/V_a	1.25
Lift-off safety factor	1.3
Governing fuel	Methane

4.2 Combustion Calculations

For a specified thermal power, the required fuel mass flow rate was calculated using the lower heating value (LHV) of the selected fuel.

$$\dot{m}_f = \frac{P}{LHV}$$

Where

- P = thermal power (W)
- LHV = lower heating value of the fuel (J/kg)

The corresponding air mass flow rate was obtained from the stoichiometric air-fuel ratio,

$$\dot{m}_a = \dot{m}_f \times AFR$$

The volumetric flow rates were then calculated as

$$Q = \frac{\dot{m}}{\rho}$$

where ρ is the density of the respective fluid.

These flow rates form the basis for sizing the fuel tube and the surrounding air annulus.

4.3 Governing Fuel Selection

The burner was required to operate with both methane and propane using a common nozzle geometry. Therefore, the governing fuel was selected based on the lift-off velocity criterion.

The lift-off velocity was estimated using the Kalghatgi-type correlation,

$$U_{LO} = 50 S_L \left(\frac{\rho_f}{\rho_a} \right)^{1.5}$$

where

- S_L is the maximum laminar burning velocity,
- ρ_f is the fuel density,
- ρ_a is the air density.

From the calculations,

Table 4.2:

Fuel	Lift-off velocity (m/s)
Methane	8.0
Propane	42.3

Since methane has the lower lift-off velocity, it represents the limiting operating condition. Therefore, the nozzle geometry was designed using methane, while propane was verified against the same burner dimensions.

The design fuel velocity was selected by applying the safety factor,

$$V_f = \frac{U_{LO}}{SF}$$

Where, in our case

$$SF = 1.3$$

Hence,

$$V_f = \frac{8.0}{1.3} = 6.18 \text{ m/s}$$

which was used throughout the burner sizing calculations.

4.4 Fuel Tube Sizing

The required fuel flow area was calculated from the continuity equation,

$$A_f = \frac{Q_f}{V_f}$$

The corresponding inner diameter was obtained as

$$D_i = \sqrt{\frac{4A_f}{\pi}}$$

Since commercial tubes are used in fabrication, the calculated diameter was snapped to the nearest available tube size.

The final selected fuel tubes are shown in Table 4.2.

Table 4.3 Fuel Tube Selection

Fuel	Required Area (mm ²)	Required ID (mm)	Selected Tube
Methane	24.67	5.60	8 mm OD × 1.20 mm wall
Propane	9.68	3.51	6 mm OD × 1.25 mm wall

Using the selected commercial tubes, the actual design fuel velocities became

- Methane: 6.19 m/s
- Propane: 6.22 m/s

which closely match the target design velocity.

4.5 Common Casing Selection

After selecting the commercial fuel tubes, a common casing diameter was determined so that both methane and propane could operate using the same burner geometry. The annular flow area available for air flow was calculated as

$$A_{ann} = \frac{\pi}{4} (D_c^2 - D_o^2)$$

where

- D_c = casing inner diameter
- D_o = fuel tube outer diameter.

The air velocity was then obtained from

$$V_a = \frac{Q_a}{A_{ann}}$$

The original burner design was based on the fuel-to-air velocity ratio, which should satisfy

$$1 \leq \frac{V_f}{V_a} \leq 1.5$$

However, because methane and propane have different densities, using the velocity ratio alone does not accurately represent the momentum carried by each fuel stream. Therefore, the Python implementation uses the mass flux ratio, defined as

$$R_m = \frac{\dot{m}_f / A_f}{\dot{m}_a / A_{ann}}$$

Using the continuity equation,

$$\dot{m} = \rho AV,$$

the above expression becomes

$$R_m = \frac{\rho_f}{\rho_a} \left(\frac{V_f}{V_a} \right)$$

Hence, the allowable velocity-ratio limits can be directly converted into mass-flux-ratio limits as

$$R_{m,\min} = \frac{\rho_f}{\rho_a} \times 1$$

$$R_{m,\max} = \frac{\rho_f}{\rho_a} \times 1.5$$

Using the fuel and air densities employed in the program, the calculated limits are in Table 4.5.

Table 4.4:

Fuel	Lower Bound	Upper Bound
Methane	0.554	0.831
Propane	1.522	2.283

The program searches through the available commercial casing diameters and selects the smallest casing whose calculated mass flux ratio lies within these limits for both fuels.

The selected casing diameter is

$$D_c = 20.9 \text{ mm}$$

The final operating conditions are in Table 4.5.

Table 4.5:

Fuel	Annulus Area (mm²)	Air Velocity (m/s)	Mass Flux Ratio	Allowable Range
Methane	292.8	4.96	0.69	0.554–0.831
Propane	314.8	4.52	2.09	1.522–2.283

Since the calculated mass flux ratios lie within the corresponding allowable limits, the selected casing diameter satisfies the design requirements for both methane and propane.

4.6 Design Verification

The completed burner geometry was verified at both 2 kW and 5 kW operating conditions.

The following parameters were evaluated.

Reynolds Number

$$Re = \frac{\rho V D}{\mu}$$

Froude Number

$$Fr = \frac{V^2}{gD}$$

Dynamic Pressure

$$\Delta P = \frac{1}{2} \rho V^2$$

Lift-off Margin

$$LO = \frac{U_{LO}}{V_f}$$

The calculated verification results are in Table 4.7.

Table 4.7:

Fuel	Power (kW)	V_f (m/s)	V_a (m/s)	Re_f	Re_a	LO Margin
Methane	2	2.48	1.98	826	1637	3.24
Methane	5	6.19	4.96	2066	4093	1.30
Propane	2	2.49	1.81	1938	1726	16.98
Propane	5	6.22	4.52	4845	4315	6.79

The verification confirmed that all operating points satisfy the design criteria. The minimum lift-off margin of 1.30 occurs for methane at 5 kW, confirming that methane is the governing fuel for the burner design.

4.7 Final Burner Dimensions

The final burner geometry obtained from the design calculations is in Table 4.8.

Table 4.8:

Parameter	Final Value
Governing fuel	Methane
Design fuel velocity	6.18 m/s
Methane tube	8 mm OD × 1.20 mm wall
Propane tube	6 mm OD × 1.25 mm wall
Common casing bore	20.9 mm
Contraction area ratio	42.25
Contraction length	125.4 mm

These dimensions were subsequently used to generate the nozzle profile presented in the following chapter.

5. Nozzle Profile Methodology

After completing the burner sizing calculations, the next step is to generate the converging nozzle profile. The purpose of the nozzle is to provide a smooth transition from the upstream burner dimensions to the burner exit while avoiding abrupt geometric changes that may disturb the flow. Unlike the burner dimensions, which are obtained from combustion and continuity calculations, the nozzle profile is generated using a mathematical curve that satisfies prescribed geometric boundary conditions.

In the present work, the nozzle profile generation methodology was developed by studying a reference MATLAB implementation. The reference code was used only to understand the mathematical formulation and the sequence of calculations. The complete methodology was then independently implemented and integrated into the Python burner design program.

5.1 Governing Nozzle Geometry

Although the burner is designed to operate with both methane and propane, only a single nozzle geometry is generated.

From the burner sizing calculations, methane was identified as the governing fuel because it has the lower lift-off velocity (8.0 m/s) compared to propane (42.3 m/s). Consequently, methane determines the limiting operating condition and governs the burner dimensions.

Therefore, the nozzle profile is generated using the methane burner geometry, while propane is verified using the same nozzle.

5.2 Exit Geometry

The nozzle exit dimensions are not assumed. They are obtained directly from the burner sizing calculations presented in Section 4.

The exit radii used in the present work are summarized in Table 5.1.

Table 5.1 Nozzle Exit Geometry

Wall	Exit Radius (mm)	Exit Diameter (mm)
Fuel inner	2.80	5.60
Fuel outer	4.00	8.00
Casing inner	10.45	20.90
Casing outer	11.90	23.80

Since these dimensions originate from the burner sizing calculations, they remain fixed during the nozzle generation process.

5.3 Inlet Geometry

The inlet dimensions of the nozzle are determined using the prescribed area contraction ratio.

In the present implementation,

$$CR = 42.25$$

which was taken from the calculation as below formula,

which corresponds to

$$\frac{A_{in}}{A_{out}} = 42.2$$

Since

$$A = \frac{\pi D^2}{4},$$

the inlet diameter is obtained from

$$D_{in} = D_{out} \sqrt{CR}$$

The calculated inlet dimensions are given in Table 5.2.

Table 5.2 Nozzle Inlet Geometry

Wall	Inlet Radius (mm)	Inlet Diameter (mm)
Fuel inner	18.20	36.40
Fuel outer	26.00	52.00
Casing inner	67.92	135.84
Casing outer	77.35	154.70

5.4 Contraction Length

The axial length of the nozzle is defined using a constant length-to-casing ratio.

The MATLAB reference nozzle has an approximate contraction length-to-exit-diameter ratio of 6.8. Based on this observation, the contraction length in the present work was defined using a constant length-to-casing ratio,

$$L = 6D_c$$

Where

$$D_c = 20.9 \text{ mm}$$

Therefore,

$$L = 6 \times 20.9 = 125.4 \text{ mm}$$

This length is used for all four nozzle walls.

5.5 Generation of the Nozzle Walls

The nozzle consists of four independent wall profiles:

- Fuel inner wall
- Fuel outer wall
- Casing inner wall
- Casing outer wall

Each wall is generated independently using a fifth-order polynomial. For every wall, the inlet radius, exit radius, contraction length, and midpoint constraint are specified. The polynomial coefficients are then calculated so that the resulting profile satisfies the required geometric conditions at both ends while maintaining a smooth converging shape.

Finally, the four wall profiles are combined to obtain the complete nozzle geometry shown in Figure 1 in Result section.

6. Mathematical Formulation

The converging nozzle profile is generated independently for each of the four nozzle walls using a fifth-order polynomial. The polynomial coefficients are determined such that the resulting profile satisfies the prescribed geometric constraints at the inlet, outlet, and midpoint of the nozzle.

6.1 Quintic Polynomial Representation

The nozzle wall profile is represented by a fifth-order polynomial,

$$r(x) = a_5x^5 + a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0$$

Where

- $r(x)$ = nozzle wall radius,
- x = axial coordinate,
- L = contraction length,
- $a_0, a_1, \dots, a_5$ = polynomial coefficients.

Since the polynomial contains six unknown coefficients, six independent boundary conditions are required.

6.2 Boundary Conditions

To ensure a smooth converging nozzle profile, the following boundary conditions are imposed.

At the nozzle inlet

Radius

$$r(0) = R_{in}$$

Zero slope

$$r'(0) = 0$$

At the nozzle exit:

Specified exit radius

$$r(L) = R_{exit}$$

Zero exit slope

$$r'(L) = 0$$

Zero exit curvature

$$r''(L) = 0$$

Midpoint constraint

To control the overall curvature of the nozzle,

$$r\left(\frac{L}{2}\right) = fR_{in}$$

Where

f is the midpoint fraction.

The midpoint fractions used in the present work are listed in Table 6.1.

Table 6.1 Midpoint Fractions

Nozzle Wall	Midpoint Fraction (f)
Fuel inner	0.54
Fuel outer	0.56
Casing inner	0.57
Casing outer	0.575

6.3 Matrix Formulation

Applying the six boundary conditions results in the following linear system,

$$Ax = b$$

Where

$$x = [a_5, a_4, a_3, a_2, a_1, a_0]^T$$

The coefficient matrix is

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ L^5 & L^4 & L^3 & L^2 & L & 1 \\ 5L^4 & 4L^3 & 3L^2 & 2L & 1 & 0 \\ 20L^3 & 12L^2 & 6L & 2 & 0 & 0 \\ \left(\frac{L}{2}\right)^5 & \left(\frac{L}{2}\right)^4 & \left(\frac{L}{2}\right)^3 & \left(\frac{L}{2}\right)^2 & \frac{L}{2} & 1 \end{bmatrix}$$

and

$$\mathbf{b} = \begin{bmatrix} R_{in} \\ 0 \\ R_{exit} \\ 0 \\ 0 \\ fR_{in} \end{bmatrix}$$

The polynomial coefficients are obtained by solving

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$$

using the numerical linear algebra routines available in Python.

6.4 Nozzle Geometry

The inlet radius of each nozzle wall is calculated from the specified contraction ratio as mentioned in section 5,

$$CR = \frac{A_{in}}{A_{out}}$$

Since

$$A = \frac{\pi D^2}{4},$$

the inlet diameter becomes

$$D_{in} = D_{out}\sqrt{CR}$$

For the present work,

$$CR = 42.25$$

and therefore

$$D_{in} = 6.5D_{out}$$

The contraction length is

$$L = 6D_c = 6 \times 20.9 = 125.4 \text{ mm}$$

7. Code Modifications

The original Python program was capable of performing burner sizing and design verification but did not include automatic nozzle profile generation or geometry export. In the present work, the code was extended by incorporating a nozzle generation module based on a fifth-order polynomial formulation. In addition, the generated nozzle profiles were exported as coordinate files so that the analytical geometry could be directly utilized for CAD modelling. The major modifications made during the implementation are summarized below.

Table 7.1 Major Code Modifications

Module	Modification	Purpose
Governing fuel selection	Methane selected automatically based on the lowest lift-off velocity	Generate a single nozzle geometry for both fuels
Nozzle generation	Added fifth-order polynomial-based nozzle profile generation	Generate smooth converging nozzle walls
Geometry calculation	Inlet radii calculated using a contraction ratio of 42.25	Scale the nozzle from burner exit dimensions
Contraction length	Adopted $L = 6D_c$ based on non-dimensional scaling with the reference geometry	Preserve similar geometric proportions
Boundary conditions	Implemented six geometric constraints for each nozzle wall	Ensure smooth inlet and outlet transitions
Profile verification	Added checks for continuous convergence and positive wall spacing	Verify the generated nozzle geometry
CSV export	Automatically exported four nozzle wall profiles as separate CSV coordinate files	Provide CAD-ready geometry for subsequent modelling
Visualization	Developed automatic plotting of the complete nozzle profile	Enable geometry visualization and comparison

The nozzle generation module was integrated with the existing burner sizing program so that the complete design process, from combustion calculations to nozzle profile generation, can now be performed automatically. The final implementation generates a single governing nozzle geometry based on methane while maintaining compatibility with both methane and propane operating conditions.

8. Results

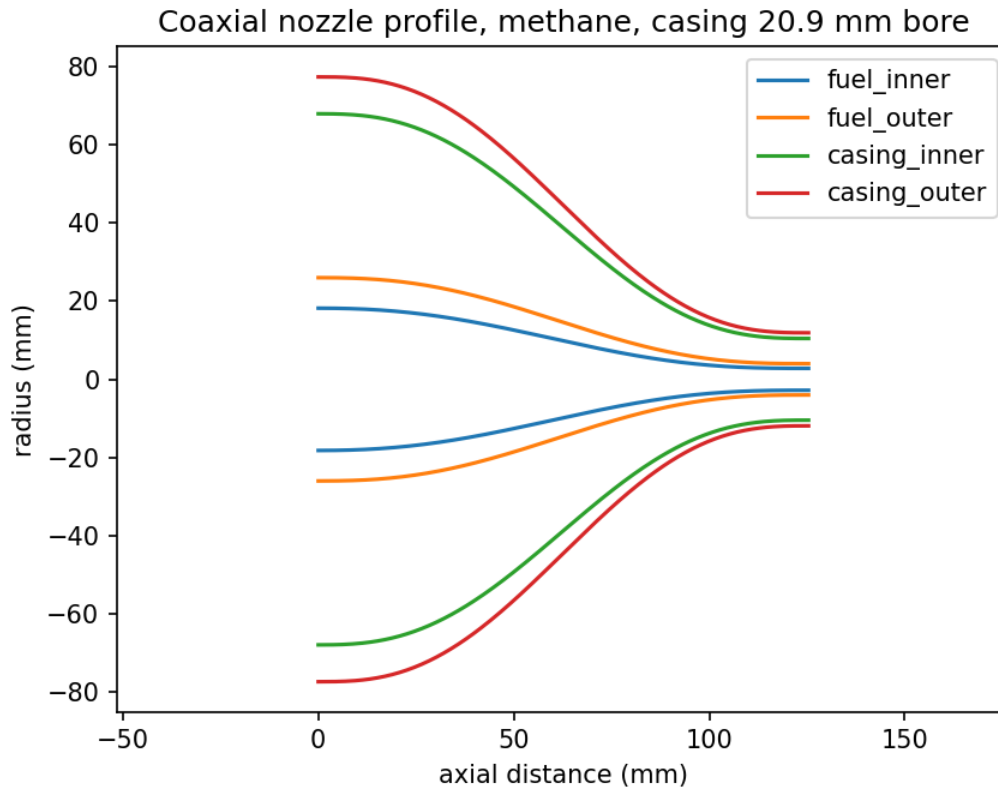


Figure 1: Final Nozzle Profile

The profile is based on the governing fuel (methane) and consists of four smoothly converging walls representing the fuel inner wall, fuel outer wall, casing inner wall, and casing outer wall.

After successful generation and verification of the nozzle profile, the program exports the coordinates of each wall as separate CSV files, namely: **InnerNozzle_InnerSurface.csv**, **InnerNozzle_OuterSurface.csv**, **OuterNozzle_InnerSurface.csv**, and **OuterNozzle_OuterSurface.csv**. Each file contains the axial (x) and radial (y) coordinates describing the corresponding nozzle wall.

These exported coordinate files preserve the analytically generated nozzle geometry and provide the geometric input required for updating the existing burner CAD model or constructing the nozzle geometry in SolidWorks without manually recreating the profile.

	InnerNozzle_InnerSurface.csv	Excel.CSV
	InnerNozzle_OuterSurface.csv	Excel.CSV
	OuterNozzle_InnerSurface.csv	Excel.CSV
	OuterNozzle_OuterSurface.csv	Excel.CSV

Figure 2: Exported Coordinate Files for CAD Geometry Generation

9. MATLAB Reference

9.1 MATLAB Code - Curve_generator.m

```
clc;
clear all;
f=2;
Y=15.3/f; %outer nozzle inlet -inner surface
Y2=9.3/f; %inner nozzle inlet -outer surface
Y3=7.8/f; % inner nozzle inlet -inner surface
Y4=16.5/f; %outer nozzle inlet -outer surface

YY=1.1; %outer nozzle outlet -inner surface
YY4=1.275;%outer nozzle outlet -outer surface

YY3=1.2/f; %inner nozzle outlet -inner surfacea
YY2=YY3+0.1;%inner nozzle outlet -outer surface

L=30/f;
L1=0.5*L;
L2=0.5*L;
L3=0.5*L;

%%%profile solver
y1=0.57*Y;
y2=0.56*Y2;
y3=0.54*Y3;
y4=0.59*Y4;

Ao=[0 0 0 0 0 1;5*L.^4 4*L.^3 3*L.^2 2*L 1 0; 20*L.^3 12*L.^2 6*L 2*L 0
0; 0 0 0 0 1 0; L.^5 L.^4 L.^3 L.^2 L 1; L1.^5 L1.^4 L1.^3 L1.^2 L1 1 ];
Bo=[Y 0 0 0 YY y1];

Ac=[0 0 0 0 0 1;5*L.^4 4*L.^3 3*L.^2 2*L 1 0; 20*L.^3 12*L.^2 6*L 2*L 0
0; 0 0 0 0 1 0; L.^5 L.^4 L.^3 L.^2 L 1; L1.^5 L1.^4 L1.^3 L1.^2 L1 1 ];
Bc=[Y2 0 0 0 YY2 y2];

Ai=[0 0 0 0 0 1;5*L.^4 4*L.^3 3*L.^2 2*L 1 0; 20*L.^3 12*L.^2 6*L 2*L 0
0; 0 0 0 0 1 0; L.^5 L.^4 L.^3 L.^2 L 1; L1.^5 L1.^4 L1.^3 L1.^2 L1 1 ];
Bi=[Y3 0 0 0 YY3 y3];

Am=[0 0 0 0 0 1;5*L.^4 4*L.^3 3*L.^2 2*L 1 0; 20*L.^3 12*L.^2 6*L 2*L 0
0; 0 0 0 0 1 0; L.^5 L.^4 L.^3 L.^2 L 1; L1.^5 L1.^4 L1.^3 L1.^2 L1 1 ];
Bm=[Y4 0 0 0 YY4 y4];

C=linsolve(Ao,Bo. ');
D=-1*C;

E=linsolve(Ac,Bc. ');
F=-1*E;

G=linsolve(Ai,Bi. ');
H=-1*G;

I=linsolve(Am,Bm. ');
J=-1*I;
```

```

##### final profile creator
for i=1:30000
    x(i)=(i-1)/1000/f;
    ym(i)=C(1)*x(i).^5 +C(2)*x(i).^4 +C(3)*x(i).^3 +C(4)*x(i).^2+ C(5)*x(i)
+ C(6);
    yn(i)=D(1)*x(i).^5 +D(2)*x(i).^4 +D(3)*x(i).^3 +D(4)*x(i).^2+ D(5)*x(i)
+ D(6);

    ym2(i)=E(1)*x(i).^5 +E(2)*x(i).^4 +E(3)*x(i).^3 +E(4)*x(i).^2+ E(5)*x(i)
+ E(6);
    yn2(i)=F(1)*x(i).^5 +F(2)*x(i).^4 +F(3)*x(i).^3 +F(4)*x(i).^2+ F(5)*x(i)
+ F(6);

    ym3(i)=G(1)*x(i).^5 +G(2)*x(i).^4 +G(3)*x(i).^3 +G(4)*x(i).^2+ G(5)*x(i)
+ G(6);
    yn3(i)=H(1)*x(i).^5 +H(2)*x(i).^4 +H(3)*x(i).^3 +H(4)*x(i).^2+ H(5)*x(i)
+ H(6);

    ym4(i)=I(1)*x(i).^5 +I(2)*x(i).^4 +I(3)*x(i).^3 +I(4)*x(i).^2+ I(5)*x(i)
+ I(6);
    yn4(i)=J(1)*x(i).^5 +J(2)*x(i).^4 +J(3)*x(i).^3 +J(4)*x(i).^2+ J(5)*x(i)
+ J(6);
end

plot(x,ym,x,yn,x,ym2,x,yn2,x,yn3,x,ym3,x,ym4,x,yn4);
xlabel('axial distance in cms');
ylabel('diameter');
axis equal;
legend (' D:30 65% at midline d:5','outer jet','D:24 55% at midline
d:2','inner jet');

%plot(x,ym,x,ym2,x,ym3,x,ym4);
ym=ym';
ym2=ym2';
ym3=ym3';
ym4=ym4';
x=x';

```

9.2 MATLAB code output plot

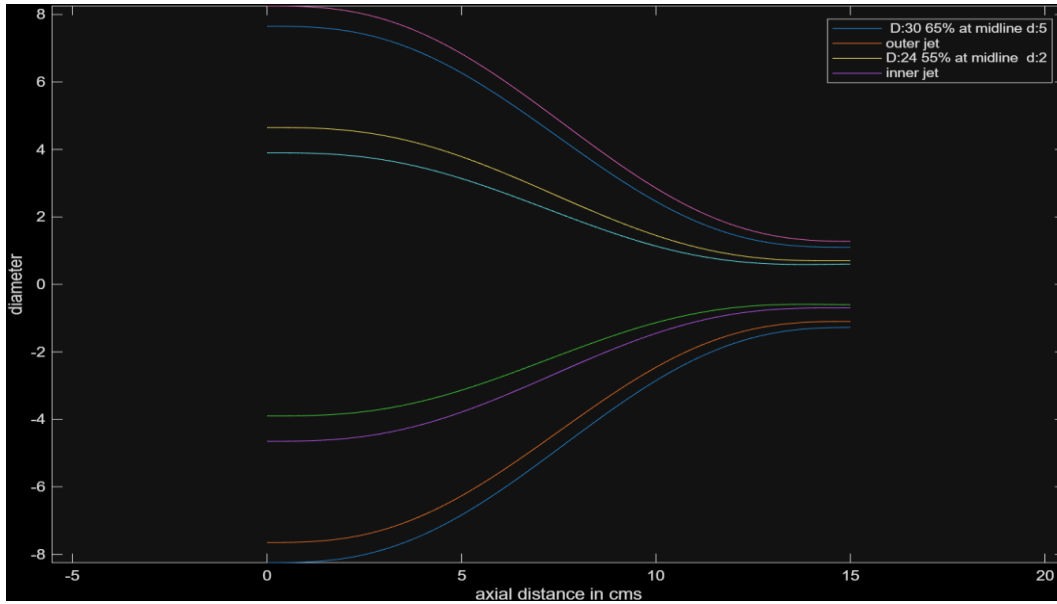


Figure 3: Reference MATLAB nozzle profile generated using a fifth-order polynomial with a contraction length of 15 cm, midpoint located at $L/2$, inlet and outlet dimensions defined in the MATLAB code, and midpoint constraints of 57%, 56%, 54%, and 59

10. References

- MATLAB reference code "Curve_generator.m" provided in section 9, provided for nozzle profile methodology reference. Used only for understanding the polynomial formulation and boundary-condition implementation; the Python implementation presented in this report was developed independently.
- Python 3.13 Documentation. *Python Standard Library*. Available at: <https://docs.python.org/3/>
- NumPy Developers. *NumPy User Guide*. Available at: <https://numpy.org/doc/>
- Kalghatgi, G. T., *Blow-Off Limits of Gaseous Jet Diffusion Flames*, Combustion Science and Technology.